

A simple water balance modelling approach for determining water availability in an irrigation tank cascade system

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Abstract

This paper presents a water balance model *Cascade* formulated to account for the dynamic hydrologic components of an irrigation tank cascade system in Anuradhapura, Sri Lanka. The model is designed to estimate tank water availability on a daily basis, for the purpose of improving productive use of water resources in the tank cascade system. It represents the physical system using a node–link system configuration, and incorporates water balance components of different types of irrigation tanks including rainfall runoff, rainfall on tank, evaporation of tank water, tank seepage and percolation, irrigation water release, spillway discharge and return flow from upstream tanks. An important feature of *Cascade* is that it employs a modified runoff coefficient method for estimating runoff from rainfall, which incorporates a modified Antecedent Precipitation Index as an indicator of catchment wetness. This provided a simplified method for representing the non-linear runoff generation process. The model calculates tank seepage and percolation based on functions derived from an analysis of the observed tank water reduction during time periods without rainfall.

The model was calibrated using field data collected at four tanks over a period of 21 months, which represented different agrometeorologic conditions encountered under both *Maha* and *Yala* growing seasons at the Thirappane tank cascade system in Anuradhapura, Sri Lanka. The model results agreed well with the measured data particularly in the two tail end tanks of the cascade, and clearly manifested the relative magnitudes of the tank water balance components and their temporal variations. An independent validation of the model could not be performed with the available data and thus the predictive capability of the model is not evaluated. The results of this study suggests that, with further improvements, the simple water balance modelling approach used in the model could potentially provide a basis for the development of a useful tool in the process of optimising usage of the limited water resources in similar tank cascade systems.

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1. Introduction

With the increasing demand for improved agricultural production to sustain growing population, the need to maximise the productive use of the limited water resources has been widely recognised. In the dry

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zone of Sri Lanka and in other similar regions, better management of water stored in irrigation tanks (reservoirs) is vital in achieving this.

Thousands of irrigation tanks have sustained agricultural production over more than 2000 years in the dry zone of Sri Lanka, providing an economical means for surface storage of runoff during the rainy season for subsequent use as irrigation water according to the requirements of rice and other crops. Many of the irrigation tanks are interconnected forming cascades, allowing surplus flow from the upstream tank(s) and return flow from the upstream command area(s) to reach the tank that is immediately downstream. This facilitates re-use of water in the command area of the downstream tank, and in effect, increases available water for irrigation. Several hundreds of such 'tank cascade systems', with small tanks in the cascade numbering from 3 to 30, have been identified in the dry zone of Sri Lanka (Tasumi et al., 1999). These occupy about 60% of the island, where rainfall is less reliable.

In the development and implementation of management practices aimed at improving the effective use of water, studies leading to the development of models that can predict available tank water in cascade systems are invaluable. It is imperative that any such models developed should be of a complexity commensurate with the generally available data, in order to provide a practical tool for improving effective water usage in tank cascade systems.

There had been several studies carried out to understand key hydrologic parameters of tank cascades and quantifying water balance components (Dharmasena, 1989, 1991; Kitamura, 1984, 1990; Itakura, 1995; Usgodaarachi et al., 1996; Shinogi et al., 1998; Tasumi et al., 1999). In attempting to explore improved management options and examine the possibility of stabilising a tank cascade through structural modifications, Itakura (1995) developed a water balance model based on a set of tanks in the Thirappane tank cascade system (TCS) in Anuradhapura, Sri Lanka (Fig. 1). In this model, runoff was estimated using a runoff coefficient that was assigned an average seasonal value, and tank seepage and percolation loss was approximated as a linear function of the tank water surface area. Shinogi et al. (1998) extended this model further by incorporating two soil-moisture parameters based on a conceptual model for

representing runoff generation, and representing tank seepage losses as a logarithmic function of the tank water surface area. The improved model was calibrated for an irrigation tank in the cascade based on a single season of hydrological measurements as described in Shinogi et al. (1998).

1.1. Objective of the study

The objective of this study was to develop a model with the capability to account for the dynamic hydrologic components of the Thirappane TCS in Anuradhapura, Sri Lanka (Fig. 1), in order to provide estimates of the tank water availability on a daily basis. The intention was to develop the model based on a simple water balance modelling approach enabling to keep the model structure as simple as possible and the data requirement to be minimal, for increasing its potential for field application. It was expected that such a model could provide the basis for the development of a useful tool, which could provide a valuable means to determine water availability in tank cascade systems such as the Thirappane TCS, which are commonly found in the dry zone of Sri Lanka and in similar regions.

2. The study area and field measurements

The Thirappane TCS, the focus of this study, is located about 20 km south of Anuradhapura in Sri Lanka (Fig. 1). There are six interlinked tanks in this TCS and the distance from the most upstream to the most downstream tank is only 8 km (Itakura, 1995). The *Cascade* model was developed considering four tanks in the TCS namely, Vendarankulama, Bulankulama, Meegassagama and Alisthana (Fig. 1), where field measurements have been collected as reported by Shinogi et al. (1998). The Bulankulama tank contained another small tank (Badugama) within its catchment, which was not functioning at the time.

Rice, the staple food of Sri Lanka, is the main irrigated crop during both *Maha* and *Yala* growing seasons in the Thirappane TCS, in common with much of the agricultural land in the North Central Province of Sri Lanka. Based on the meteorological data from the Maha Illuppallamma Agricultural Research Station,

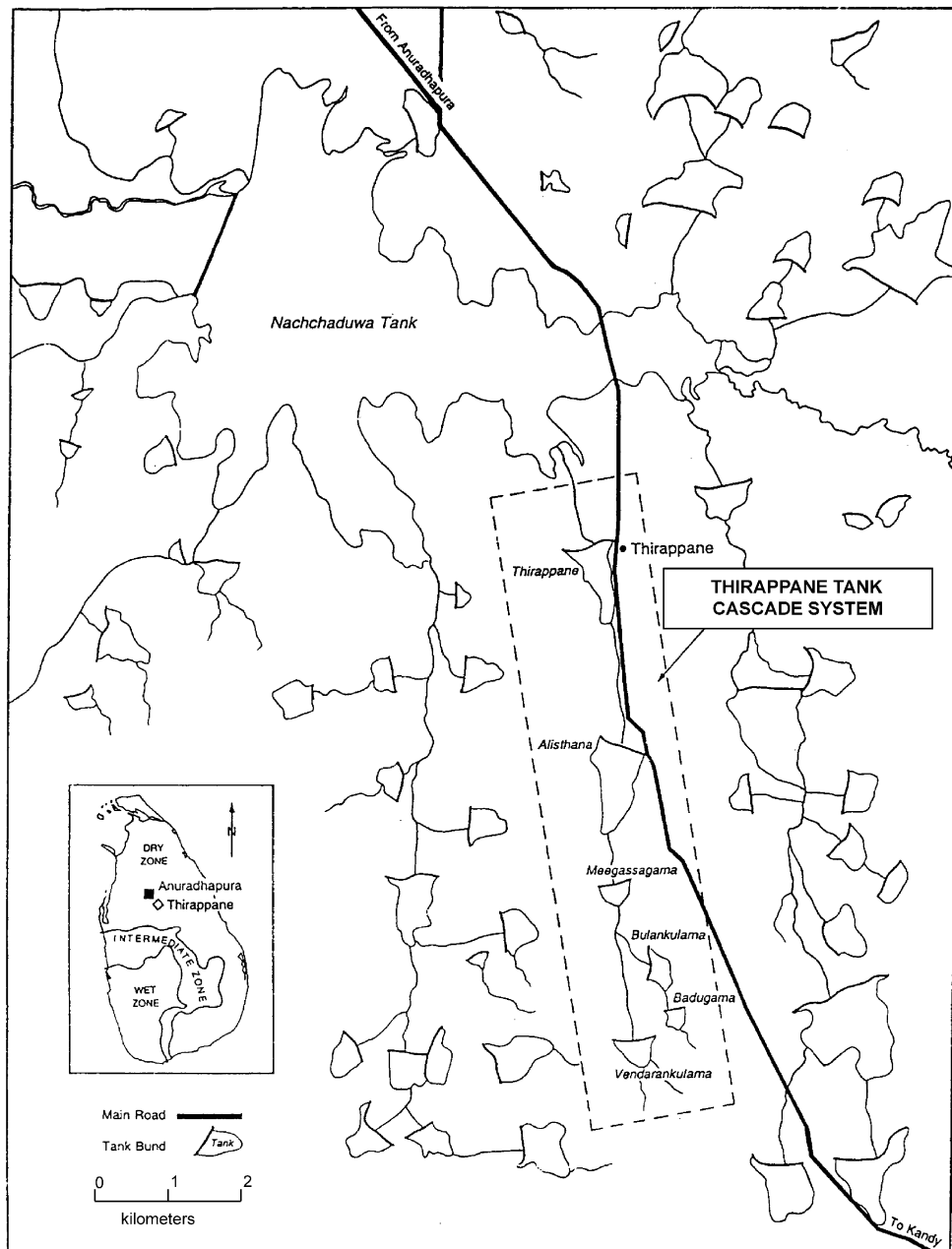


Fig. 1. Location of the Thirappane TCS.

located about 13 km from the site, the average annual rainfall of this area is about 1490 mm, and the average annual potential evapotranspiration is estimated to be much higher, 2453 mm (Shinogi et al., 1998). Except during three months (October–December) of the year,

average monthly potential evapotranspiration exceeds average monthly rainfall.

The field measurements to obtain important hydrological and physical characteristics of the Thirappane TCS have been carried out based

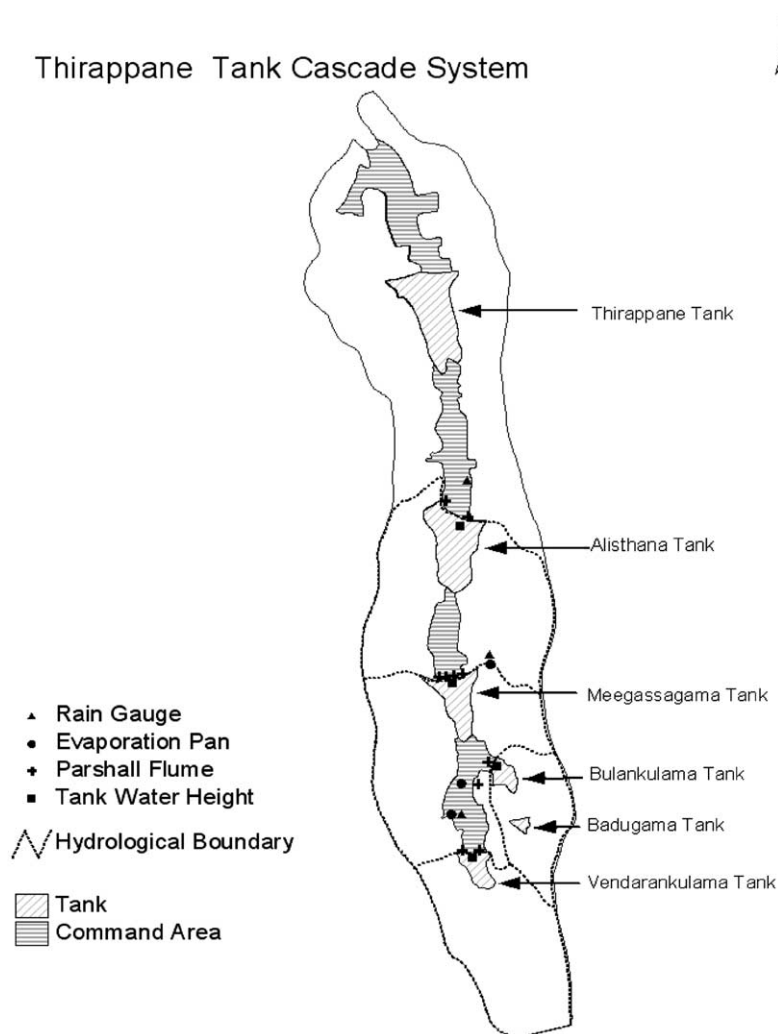


Fig. 2. The study area and locations of field measurements.

on an observation network since 1997 (Shinogi et al., 1998). These measurements have provided vital observed data required for this study on daily rainfall and pan evaporation, tank water height, and water issue for irrigation. Initially, this study utilised the observed data recorded from 22 July 1997 to 21 February 1998, and later as more data became available records until 18 April 1999 was used. However, the data sets obtained at the head-end tanks Vendarankulama and Bulankulama were not complete as rehabilitation work was carried out. Except during the rehabilitation work, the tank water levels were observed twice a day, manually in

all four tanks considered in this study. The volume of water released for irrigation was measured at Parshall flumes located downstream of the tank sluices. The daily rainfall measurements were made at each tank with the use of manually recorded rain gauges and the pan evaporation data collected at the Meegassagama tank was used in this study. The points of field measurements at each tank in the TCS are shown in Fig. 2.

Field measurements also included soil surveys, which allowed classification of the soil series. The main soil types commonly found in the North Central Province, Reddish Brown Earth and Low Humic Gley

Table 1
Thirappane TCS: catchment area description

Tank	Catchment area (m ²)	Average slope (%)		Average thickness of the aquifer (m)		
		N–S	E–W			
<i>Catchment area, average slope and aquifer thickness</i>						
Vendaramkulama	2.40 × 10 ⁶	2.97	0.36	8.2		
Bulankulama	1.13 × 10 ⁶	0.30	0.36	5.0		
Meegassagama	3.33 × 10 ⁶	0.69	0.41	5.0		
Alisthana	3.25 × 10 ⁶	0.85	0.91	6.0		
Tank	Cropped land ^a	Dense forest	Scrub land	Degraded forest	Barren land	Teak plantation
<i>Land use (% area)</i>						
Vendaramkulama	2	40	6	0	5	47
Bulankulama	6	57	8	7	11	9
Meegassagama	27	40	16	11	1	5
Alisthana	16	50	11	20	4	0
Tank	Well drained, reddish brown earth soils	Moderately to imperfectly drained reddish brown earth soils	Poorly drained low Humic Gley soils	Rock outcrops		
<i>Soil type (%area)</i>						
Vendaramkulama	62	22	10	6		
Bulankulama	63	15	8	13		
Meegassagama	67	16	17	1		
Alisthana	78	9	9	4		

^a Includes paddy land, homestead gardens, chena land.

cover most of the area in the Thirappane TCS (Table 1). Land use types and physical characteristics of the catchment areas are also given in Table 1.

3. Development of the model

The *Cascade* model was formulated based on a simple structure, incorporating dynamic hydrologic processes associated with a set of four tanks in the Thirappane TCS. The configuration of the tank cascade system is represented in the model using a commonly accepted node–link system configuration that can delineate relative positions of tanks and their interconnections (Fig. 3). The nodes indicate tanks and the links indicate the interconnection between tanks.

As the initial step, each tank should be assigned a *node number* and *tank type* depending on its relative position within the cascade. Three types of tanks are identified: Start Tank (ST)—a tank with no inflow

from upstream tanks; Normal Tank (NT)—a tank with inflow from one upstream tank; and Confluence Tank (CT)—a tank with inflow from more than one upstream tank. In the Thirappane TCS, Vendaramkulama and Bulankulama have been identified as STs, and Meegassagama and Alisthana were assigned as CT and NT, respectively (Fig. 3). The node numbering begins with STs in the upstream end of the cascade and continues sequentially towards the downstream end. The CTs should always have a node number higher than those of the upstream tanks linked to them. The NT should be assigned a node number greater than that of the tank located immediately upstream. The last node number in the cascade should be equal to the number of tanks in the cascade.

3.1. Tank water balance components

The effective water balance components of the tank cascade system simulated by the *Cascade* model are shown in Fig. 4. In ST type tanks, which are

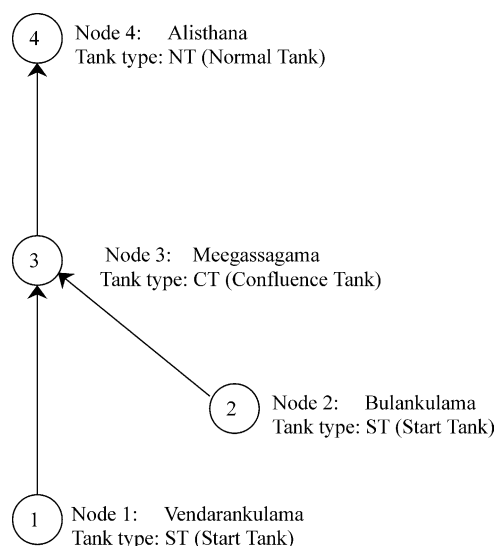


Fig. 3. Representation of the Thirappane TCS in the model.

normally located in the upstream end of the cascade, runoff from the catchment, and rainfall on the tank water surface, form the inflow components. Losses and water flow from the tank denoted as ‘outflow’ components include: evaporation of tank water, seepage through the tank embankment and percolation through the tank bed, which is referred to as

‘seepage loss’ in the model, water issue for irrigation, and spillway discharge.

In NT and CT type tanks, in addition to the runoff generated in their own catchments and the rainfall on tank water surface, fraction of the outflow from the immediately upstream tank(s) can flow in, increasing the total available tank water (Fig. 4). This additional inflow is treated as two different components: return flow due to seepage loss and water issue, and return flow due to spillway discharge. In NT tanks these flow components occur from the immediately upstream tank, whereas as in CT tanks inflow from the immediately upstream tanks that are linked to the CT tank need to be incorporated. Outflow components of NT and CT type tanks are similar to those of ST tanks. In all three types of tanks, a fraction of outflow can only reach the immediately downstream tank as return flow.

The relevant information on the node–link system configuration needs to be specified as part of the input to the model that describes the physical system. Based on the survey data at each tank, mathematical expressions for representing (1) tank area, (2) tank water volume as functions of the tank water height, and (3) tank water height as a function of tank water volume need to be formulated and incorporated to

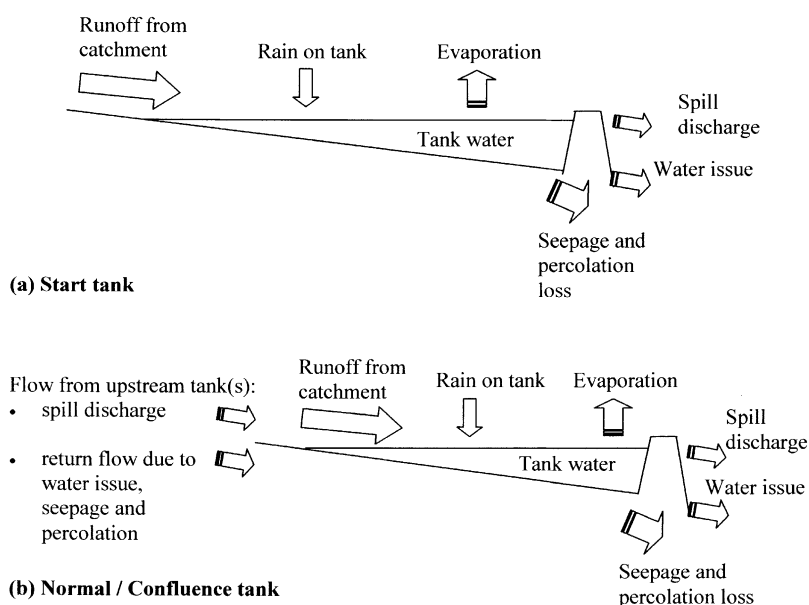


Fig. 4. Schematic of the tank water balance components.

the model. The cubic expressions derived for the Thirappane TCS depicting above relationships are given in Jayatilaka et al. (2000).

3.2. Simulation of daily tank water balance

The components of the tank water balance represent complex hydrological processes associated with the tank cascade system. For example, generation of runoff from rainfall on the catchment is a complex process involving interaction of the effective climatic, geologic and topographic factors, vegetation characteristics and the antecedent conditions of the catchment. Accurate simulation of these complex processes can impose a substantial input data requirement and a detailed mathematical representation of the physical system and the relevant processes in the model. The *Cascade* model approximates the physical system and the effective hydrologic components with the use of several expressions and assumptions described as follows.

3.2.1. Inflow components

Runoff from rainfall on catchment. The *Cascade* model estimates runoff from rainfall on the catchment of each tank on a daily basis by adopting a modified runoff coefficient method, which allows the runoff coefficient to vary daily depending on a modified Antecedent Precipitation Index (API), as described by Eqs. (1) and (1a). On a given day, runoff contribution to the tank denoted as node j is expressed as:

$$RO_j = RCF_j R_j CA_j / API_j \quad (1)$$

where RO_j is runoff yield (m^3/day), RCF_j is runoff coefficient, R_j is rainfall on the catchment (m/day), CA_j is the catchment area (m^2), and API_j is API. API, at node j depends on the number of days since the last day with rainfall (n). Depending on n , API_j can be estimated as:

$$API_j = \sum_{k=0}^{k=n} 1/(k+1) \quad (1a)$$

This method provides a means to account for both (1) the effect of field conditions more favourable for runoff generation when there has been rainfall raising soil moisture levels, and (2) the effect of decreasing soil moisture, which would gradually decrease

the runoff yield. The effect of decreasing soil moisture on runoff yield would decrease as the number of days without rainfall increases. Similarly, the increase in API, thus its effect on runoff yield, decreases with the increasing number of days without rainfall once that exceeds a certain limit. In the model this limit is taken as 11 days, and as such, when the number of days without rainfall is greater than 11 days ($n > 11$), API corresponding to $n = 11$ is used.

The above process allows estimation of runoff from rainfall on a daily basis under normal conditions. However, when there are prolonged dry spells without rainfall, soil moisture may decrease to a very low level, and runoff may not be produced in response to rainfall until the soil moisture is replenished. Particularly at the end of the *Yala* season, during the months of July and August, such prolonged dry spells can occur and the tanks may dry up completely. This condition can cause a delay in the generation of runoff when rainfall arrives, as the soil moisture level has to rise due to recuperated rainfall infiltration before runoff can occur. The model accounts for the above condition by using a parameter named *delay*, which is assigned to each node (tank) to act as an indicator of the initial loss of rainfall before runoff can occur in its catchment after a long dry spell. In such conditions, the model would not allow generation of runoff, until the cumulative rainfall since a prolonged dry spell exceeds the value of *delay* (given in mm). If a simulation starts following a long dry period, which has caused tanks in the cascade to dry up, this condition is assumed at the beginning of the simulation. In addition, if the number of consecutive days (1) without rainfall, and (2) tank being dry, exceeds a set 'upper limit', the model activates the condition related to long dry spells so that runoff generation will be delayed until rainfall exceeds the initial loss indicated by the *delay* parameter. The upper limit is assumed as 50 days, limiting the effect of the *delay* in generating runoff from rainfall only for periods that follow extended dry spells.

For the soil condition and depth prevailing in this region, the parameters *RCF* and *delay*, which are dependent on catchment characteristics such as soil type, vegetation, land slope, etc. are generally expected to be in the range: *delay*, from 0 to 300 mm; and *RCF*, from 0 to 0.35. This range can

be used in assigning values for the parameters at the start of a simulation.

Rain on tank water surface. Rainfall on the tank waterspread area is calculated as:

$$RT_j = TA_j R_j \quad (2)$$

where RT_j is rainfall on tank (m^3/day), R_j is the daily rainfall (m/day), and TA_j is the tank waterspread area (m^2), which is updated based on the latest tank water height at the beginning of each day.

3.2.2. Outflow components

Evaporation loss. The loss of tank water due to evaporation is computed as:

$$EV_j = fp E_j TA_j \quad (3)$$

where EV_j is evaporation loss (m^3/day), fp is pan coefficient for converting pan evaporation to tank water evaporation, E_j is pan evaporation (m/day) and TA_j is tank waterspread area (m^2). fp is a user-defined coefficient set to 0.8 to be representative of the average conditions in the region.

Seepage and percolation loss. The seepage and percolation of tank water through the tank bed and the embankment (referred to as the seepage loss) depends on the hydraulic conductivity of the tank bed and the embankment, and the difference between the tank water level and the water table in the surrounding area. The estimation of the tank seepage loss requires field measurements on the spatially varying hydraulic conductivity of the tank bed and flanks, and the time-varying tank water height and groundwater levels.

In the *Cascade* model, the tank seepage loss was estimated using the observed data from 22 July 1997 to 21 February 1998, the data set that was initially available for the study. The decrease in tank storage during time periods without rainfall can be estimated on a daily basis from the difference between the observed tank storage decrease (which can be estimated from the tank water height measurements) and the total of evaporation loss and the volume of water issued for irrigation. The detailed analysis performed using the estimated seepage loss at each tank indicated that the seepage loss as a percentage of tank water volume (%SP) is relatively high when the tank water height is low when compared with that in high tank water conditions (Jayatilaka et al.,

2000). This trend was commonly observed in all four tanks considered. The trend line obtained represents variation in SP percent as a log function of the tank water height and can be expressed as follows:

$$SP_j = [a_j \ln(h_j) + b_j] TV_j / 100 \quad (4)$$

where SP_j is seepage loss (m^3/day) at tank j , TV_j is tank water volume (m^3), h_j is tank water height (m), and a_j , b_j are parameters of the seepage function given in Jayatilaka et al. (2000). The *Cascade* model uses the above equation for estimating tank seepage loss. At the start of the simulation, TV_j is determined using the cubic expression representing tank water volume as a function of tank height given in Jayatilaka et al. (2000). Subsequently, TV_j is estimated at the end of each day, as part of the water balance calculations performed by the model.

In many field situations the required data to determine parameters of the seepage function relevant to each tank may not be available. The analysis performed in this study has been extended to derive a function that can approximate the water loss due to seepage in similar tank cascade systems. The seepage functions of the Thirappane TCS can be presented based on a common horizontal-axis representing 'Relative tank water height' defined as 'Tank water height/Tank water height at the full supply level'. Using the average of the seepage rates of tanks given by the approximated (model) relationship at different relative tank water heights, a log function indicating variations of seepage rates at the tank cascade in general can be derived as presented in Fig. 5 (first graph). Fig. 5 also includes comparisons of the derived Log(Ave) function with the observed (estimated) seepage rates (%SP) at each tank. This function can provide a useful means for estimating seepage losses in irrigation tank cascades such as the Thirappane TCS when the required information to determine the relevant parameters of the seepage function is unavailable.

Water issue for irrigation. The water released for irrigation (WQ_j) is a daily field measurement required as part of the input when calibrating the model.

Spillway discharge. When the tank water height is greater than the weir crest level, initially the spillway discharge is computed assuming it can be approximated with Eq. (5), which is applicable for broad

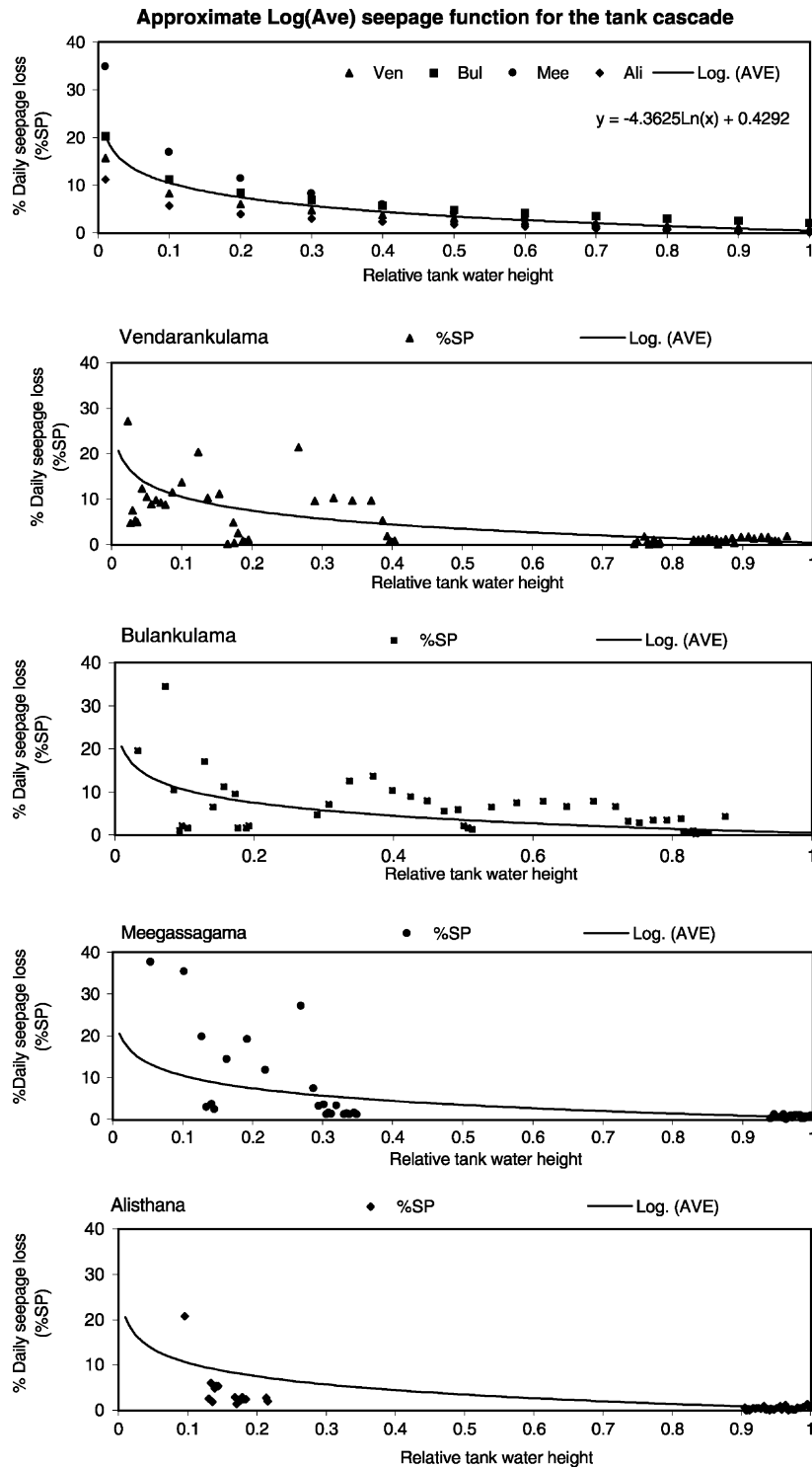


Fig. 5. Percent tank seepage loss as a function of the relative tank water height.

crested weirs (French, 1994):

$$SL_j = fdL_j(h_j - V_j)^{1.5} \quad (5)$$

where SL_j is spillway discharge (m^3/s), L_j is the length of the spillway (m), V_j is the spill level (m), and h_j is the tank water height (m) and fd is the discharge coefficient taken as $1.7 \text{ m}^{1/2}/\text{s}$ (French, 1994). SL_j is then converted to a daily rate to obtain the upper limit of the spillway discharge allowed by the model on a given day. The actual discharge is determined by the lower of the following two quantities: (a) estimation based on Eq. (5), and (b) tank water volume in excess of the full supply level (tank capacity at the spill level).

3.2.3. Additional inflow components

Return flow from upstream tanks, which forms additional inflow components to NT and CT tanks are estimated as shown.

Return flow due to seepage loss and water issue. During *Maha*, return flow due to seepage loss and water issue from upstream tanks is estimated from Eq. (6):

$$RF_j = \sum_{k=m}^n [fr(WQ_k + SP_k)] \quad (6)$$

where RF_j is return flow at tank j (m^3/day), k is the contributing upstream node number, m is the first contributing upstream node, n is the last contributing upstream node, WQ_k is water issue at the upstream node k (m^3/day), SP_k is the seepage loss at upstream node k (m^3/day), fr is a user-defined coefficient ($0 \leq fr \leq 1$).

It is assumed that the water released for irrigation during the *Yala* season would be just adequate to provide for the requirements in the command area, and as such, part of the irrigation release would not arrive as return flow at the downstream tank. The return flow component in *Yala* is therefore, estimated from Eq. (6) with WQ set to zero.

fr denotes the fraction of the seepage and water issue in the upstream tank that would become available as return flow at the downstream tank. Ponrajah (1984) noted that 20% of the irrigation release for cultivation in the immediately upstream tank is available as drainage for re-use in the downstream tank.

It is difficult to assign a return flow fraction that would be applicable for both flow components

(seepage and irrigation release) in each tank, under the spatially and temporally varying conditions in a cascade system. On the other hand it is not feasible to determine different return flow fractions for the two components (seepage and irrigation release) at each tank in the cascade. The coefficient fr is expected to provide a reasonable estimate of the return flow fraction, which would be generally applicable to the tank cascade. Initially fr can be set to be in the range 0.1–0.4, and this can be refined through calibration of the model.

Return flow due to spillway discharge. The return flow due to spillway discharge at upstream tanks is estimated as:

$$SI_j = \sum_{k=m}^n (fsSO_k) \quad (7)$$

where k is contributing upstream node number, m is the first contributing upstream node, n is the last contributing upstream node. SI_j is inflow at tank j due to spillage at upstream tank (m^3/day), SO_k is spill discharge at upstream node k (m^3/day), and fs is a user-defined coefficient which determines the fraction of spill discharge that arrives at the downstream tank. Shinogi et al. (1998) used 0.57 and 0.67 for fs at the Vendarankulama and Bulankulama tanks, respectively. Although in general fs could vary from tank to tank, in the model it is assumed to be a common coefficient and can be set to a value in the range, $0 \leq fs \leq 1$. In the absence of field measurements to determine fs , it is assumed that 0.5 would be a reasonable value.

In estimating return flow, it is assumed that part of the water released for irrigation, tank seepage loss and spillway discharge from the upstream tank can arrive at the downstream tank within the same day. As the model is designed for cascades with small irrigation tanks with relatively small command areas this assumption is considered to be reasonable.

3.2.4. Processing procedure

The daily water balance for tank j in the cascade can be written in the form of Eq. (8).

$$I_j - O_j = DS_j \quad (8)$$

where I_j is total inflow, O_j is outflow and DS_j is the change in storage determined from the difference

between tank storage at the beginning and the end of the day.

Starting from the most upstream end of the tank cascade, the model performs water balance computations for each tank on a daily basis. Tank storage at the end of the day is taken as the storage at the beginning of the next day. Inflow components and outflow components of the tank water balance are substituted to Eq. (8) in the order explained in the processing procedure illustrated by the flowchart presented in Fig. 6. It provided the basis for the *Cascade* model code, which is written as a *Lahey Fortran* program.

3.3. Model input and output

The model input includes: meteorological data (daily rainfall and pan evaporation), information on the tank cascade system (for each node: node number, tank type, spill level, RCF, *delay*, initial tank water height), user-defined coefficients (*fp*, *fr*, and *fs* set as described in Section 3.2). In the process of calibration of the model, water released for irrigation measured in the field is used as model input. In using the model for predictions, the required water release needs to be estimated considering land preparation water requirements and crop water requirements during the growing stage, based on cultivation extents of different crop types in the command area. In either case, through the daily water balance computations, the model allows release of the required irrigation water when the storage is adequate. If the storage is insufficient, the model calculates the allowable water release, and it will be less than the measured water release for the day (when calibrating the model) or the estimated water release requirement of the day (during predictions). In addition to the allowable water issue, the model provides output at the end of each day including the tank water height, tank storage, and information on the tank water balance components.

4. Model calibration

4.1. Model simulations

The *Cascade* model was calibrated in two stages using daily field measurements at four tanks of

the Thirappane TCS, collected by Shinogi et al. (1998). The initial calibration (Stage 1) was from 22 July 1997 to 21 February 1998. In the second stage (Stage 2), calibration was extended until 18 April 1999. The extended data set spanned over two *Maha* and one *Yala* growing seasons, thus represented both dry and wet condition in the tank cascade system. Initial condition in the tank cascade was assigned based on the measured tank water levels at 8 am on 22 July 1997. The parameters of the seepage functions used in the model, which have been derived based on the relevant information from the data set used in Stage 1, kept unchanged as the model calibration was extended in Stage 2. The user-defined coefficients (*fp*, *fr* and *fs*) and the model parameters RCF and *delay* were initially assigned values as recommended in Section 3 and were adjusted during model calibration when necessary. The calibration was finalised in Stage 2 involving 21-months of field data, based on a qualitative comparison of the match between the observed and simulated tank water heights over the calibration period. The use of a numerical criterion in this process would not have been appropriate, as model simulations cannot be directly compared with the measured data during the periods of rehabilitation work carried out at the two head-end tanks in the cascade. The model input, user-defined coefficients and the parameters of the calibrated model are given in Table 2.

4.1.1. Results and discussion

The model performed daily water balance computations for each tank in the cascade and calculated water balance components and the tank water height at the end of each day over the calibration period. The simulated tank water height at the end of each day was compared with the tank water height measurements taken at the beginning of the next day. The comparisons of the observed data and the simulated results at all tanks are given in Jayatilaka et al. (2001), and this paper includes the comparisons at Vendarankulama (Fig. 7) and Alisthana (Fig. 8), representing the type of results obtained in head–head and tail-end tanks, respectively. Each figure includes (1) rainfall and evaporation data, (2) observed and predicted water height, (3) inflow components, (4) outflow components, and (5) observed tank water issue along

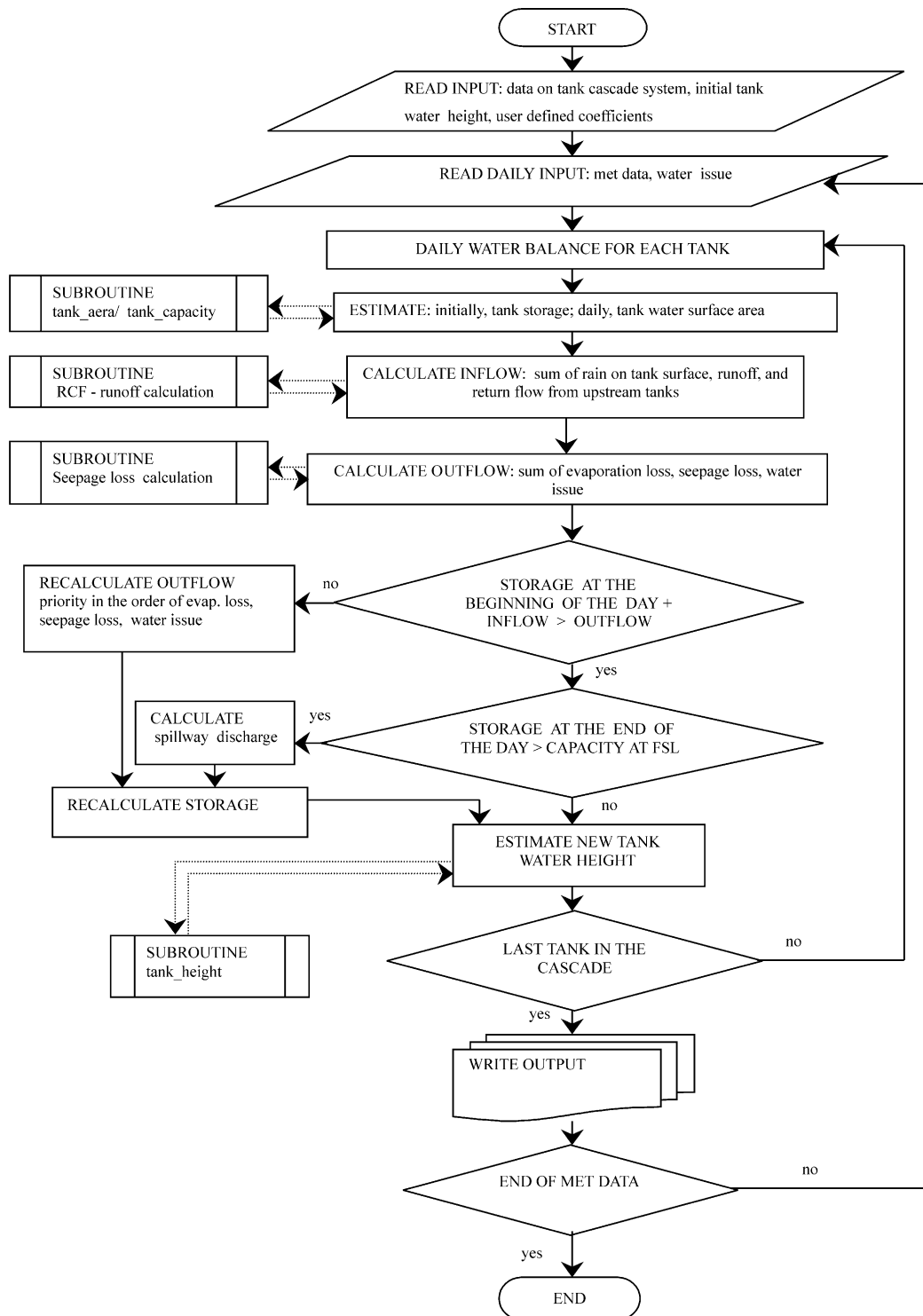
Fig. 6. Flowchart of the *Cascade* model.

Table 2
Model calibration: input parameters

<i>fp</i>	<i>fr</i>	<i>fs</i>					
<i>User-defined coefficients</i>							
0.8	0.10	0.5					
Tank	Node number	Tank type	Effective spill level (m)	Weir crest length (m)	Initial tank water height (m)	RCF	Delay (mm)
<i>Input data on tanks</i>							
Vendarankulama	1	ST (start type)	3.00	30.0	1.21	0.21	80.0
Bulankulama	2	ST (start type)	2.50	30.0	0.48	0.30	290.0
Meegassagama	3	CT (confluence type)	2.75	55.0	0.97	0.132	240.0
Alisthana	4	NT (normal type)	3.75	30.0	0.81	0.31	260.0

with the water issue allowed by the model, over the of 21-months considered.

Except for the apparent differences in the latter part of the simulation period at the head-end tanks, a reasonable overall agreement was obtained between the simulated results and the measured as shown in Figs. 7 and 8 and by Jayatilaka et al. (2001). However, a close match between the observed data and model simulations is an indication of the combined accuracy in simulating the different hydrologic components by the model. For example, if rainfall runoff is underestimated while the tank seepage loss is also underestimated the simulated tank water height may still show a close agreement with the observed data. Therefore, it is important that whenever possible, validity of the simulated results should be verified considering field evidence and understanding of the physical system.

The model simulations started on 22 July 1997 when the rainfall in the *Yala* season has ended, and dry conditions prevailed in the tank cascade system until rainfall arrived in mid September. The model simulation over the initial dry period agreed well with the field observations. The generation of runoff in response to rainfall following the dry period was delayed due to depleted soil moisture levels in the catchments at the end of the long dry spell. The model accounted for the delay in runoff generation in catchments by adjusting the *delay* parameter in the runoff calculation procedure. During the 1997/1998 *Maha* season which followed the dry spell, the simulated tank water height showed a close agreement with the observed data, particularly in the early part of

the rainfall period. Towards the end of the *Maha* season rainfall period, the model underestimated the tank water height at head-end tanks. The agreement of the model simulations with field observations over the rest of the calibration period was considerably better in the tail-end tanks when compared with the head-end tanks.

At the head-end tanks, a direct comparison of the simulated and measured tank water height is not possible over the period during and after the rehabilitation work. Vendarankulama was emptied on the 26 June 1998 for rehabilitation work and the tank remained dry until the end of December 1998. As the water issue data was not supplied to the model over this period, the model did not allow any water release, but simulated an increase in tank storage in response to rainfall. The reasons for the apparent discrepancy between the observed and the measured data since the end of the rehabilitation work are not clear. It is possible that the changes resulting from the rehabilitation work were not adequately represented in the model. Also, inconsistency in the field data used in formulating expressions such as the tank height versus tank capacity relationship cannot be ruled out.

As the 21-months of data available was used to complete the calibration stages 1 and 2, an independent validation of the model could not be performed. As a result, the predictive capability of the model cannot be assessed. In the following sections, the applicability of the methods employed by the model to account for the dynamic hydrologic components of the tank cascade system is analysed considering the model simulations, field observations and

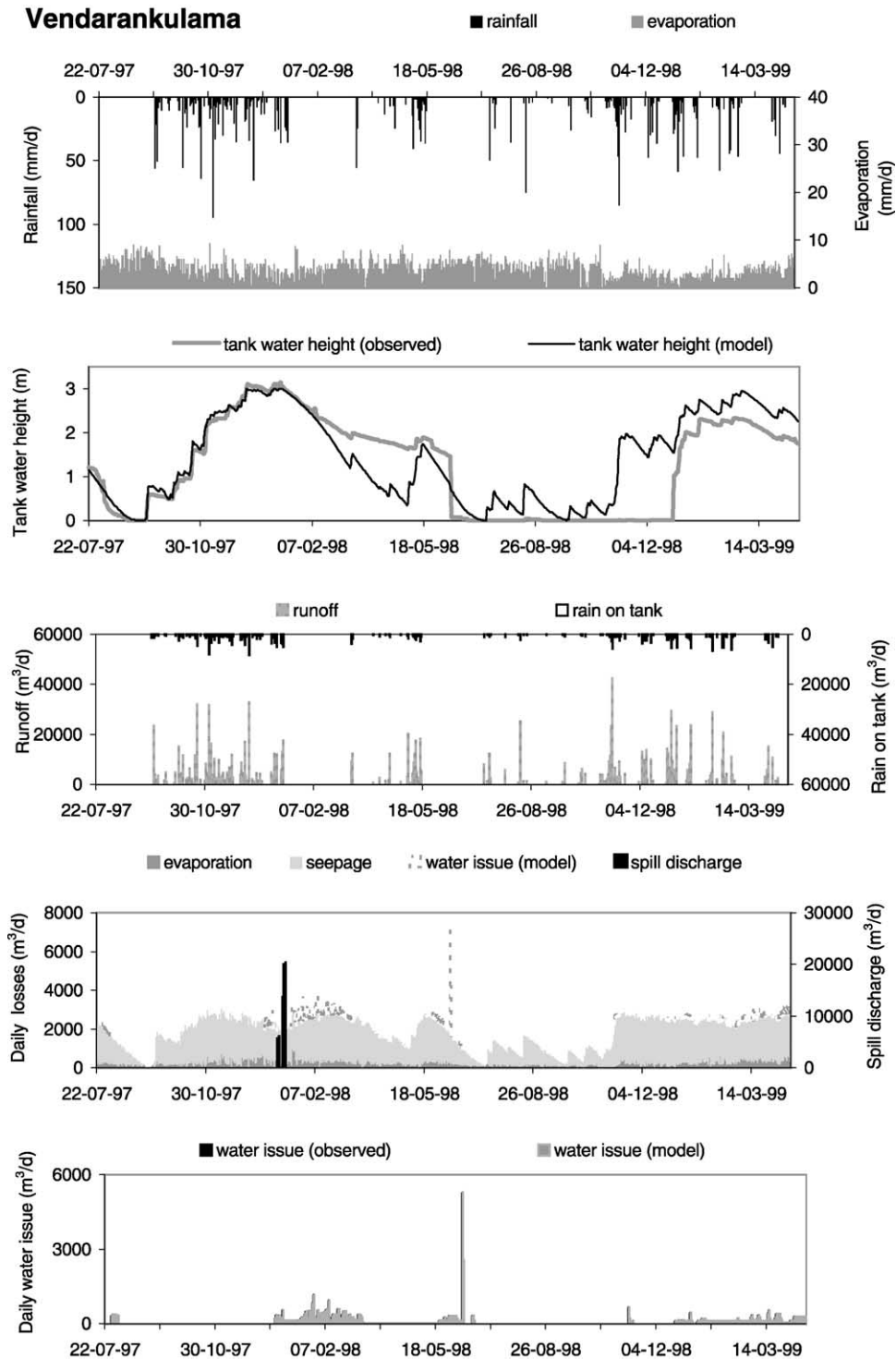


Fig. 7. Model calibration: observed and simulated tank water height, volume and simulated tank water balance components—Vendarankulama.

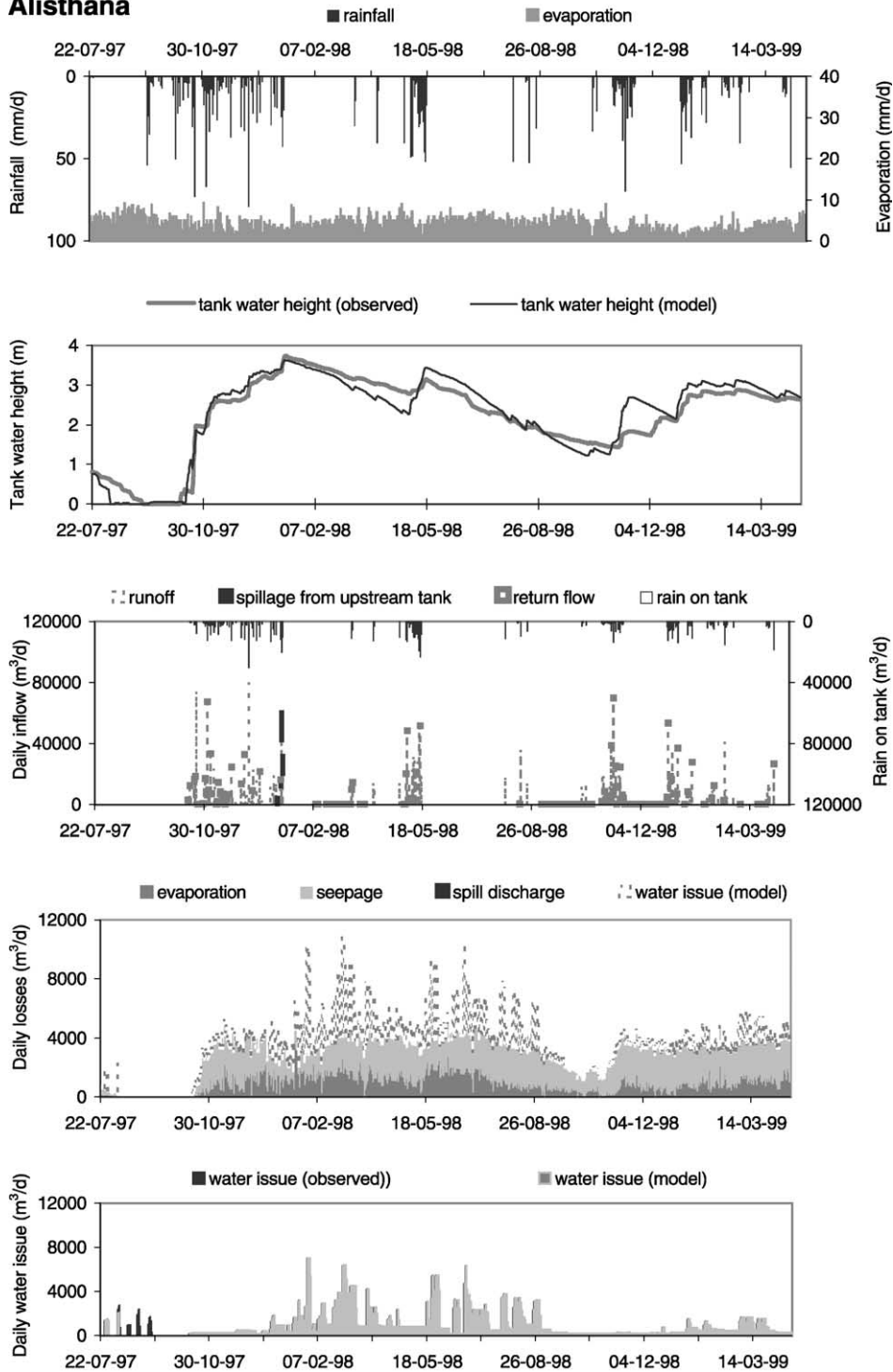
Alisthana

Fig. 8. Model calibration: observed and simulated tank water height, volume and simulated tank water balance components—Alisthana.

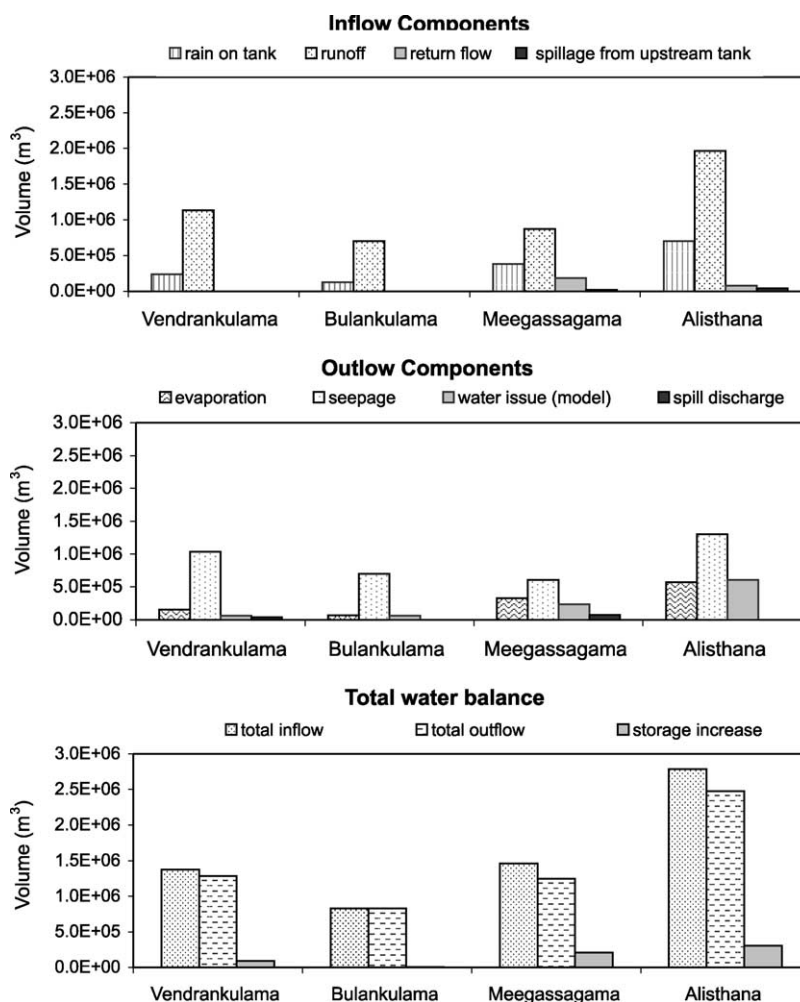


Fig. 9. Total water balance of the tank cascade over the model calibration period.

the simulated tank water balance components over the calibration period (Fig. 9).

Rainfall runoff simulation. The simulated tank water level increase in response to rainfall events over the calibration period generally agreed well with field observations, providing confidence in the method adopted in simulating runoff yield from rainfall. However, differences were apparent in the RCF (Runoff coefficient) and *delay* parameter values used in the model for simulating runoff in the tank cascade (Table 2), which may have resulted from the influence of the factors affecting runoff generation such as vegetation (land use), catchment slope and soil type. On the other hand,

incorrect data, intercorrelation of parameters and the empirical nature of the equations used in the model could also have contributed to these differences.

At the Vendarankulama tank, a low *delay* parameter was used compared with the range of the values used (Table 2). This tank is located at the foot of a hill in the upstream end of the cascade, and its catchment area is relatively steep and has a smaller area of teak plantations in comparison with other catchments in the cascade (Table 1). It can be postulated that the larger teak plantations areas, smaller cropped area, and the erosion pattern observed in this relatively steep catchment could have

influenced both RCF and *delay* parameters, as discussed by Dharmasena (1994) in studying the effect of land use on rainfall runoff relationships and runoff threshold values. On the other hand, the RCF and *delay* parameter values used could have contributed to the discrepancy observed in model simulations during February–June and October–November in 1998, and the model simulations could have been improved by increasing the RCF and *delay* parameters.

The catchment area of the Bulankulama tank contains a larger area of rock outcrops compared with the other catchments (Table 1). This could promote runoff generation, and thus, would have influenced the RCF values used at this tank, which is towards the higher end of the range of the values used in the cascade (Table 2). In the catchment area of the Meegassagama tank, cropped area is relatively large and rock outcrops represent a small area (Table 1). The RCF value used at the Meegassagama tank is the lowest in the range of values used. In addition to the influence from the factors affecting runoff generation in this catchment, there can be other plausible explanations for the low RCF value. It is possible that a low RCF was required to compensate for a possible overestimation of the area that is effective in contributing to the Meegassagama tank, or an inaccuracy in estimating tank capacity due to an inconsistency in field data.

Spillway discharge. The model simulated spillway discharge at the Vendarankulama and Meegassagama tanks (Jayatilaka et al., 2001), and this agreed well with the field observations. However, in all the tanks except Alisthana, the match between the observed data and the model was obtained with the use of weir crest levels slightly different from those indicated by the survey data. For example, at Vendarankulama a weir crest level of 3.0 m was used in the model, slightly higher than the crest level indicated by the survey data (2.9 m). It can be envisaged that field practices such as the use of sandbags to raise the spill level, and in some cases, extension of the sections of the spillway, and other field conditions could have contributed to the changes in the effective spill crest level. In addition, it is not clear whether the survey data used in this study represent the existing physical conditions accurately.

Tank water seepage. The parameters of the seepage functions derived using the data used in

calibration Stage 1 were used with out any changes in Stage 2. The overall match between the model results and the observed data in Stage 2 provided a degree of confidence in the tank seepage estimation procedure adopted in the model.

It is apparent, however, at times the model underestimated the tank storage. For example, at the Meegassagama tank, the overall match between the model results and the field measurements was considerably close; however, over the relatively dry period from mid June to November 1998, the model simulations deviated from the measured data (Jayatilaka et al., 2001). At least in part, this discrepancy may have resulted from a possible underestimation of the seepage rates by the model. In particular, the seepage function at the Meegassagama tank need to be refined to provide more realistic estimates under the high storage conditions involving more observed data under high tank water conditions.

Water issue for irrigation. The model allowed water issue for irrigation as indicated by the measured data over the simulation period except over short periods when the model indicated drying up of the tank (e.g. in August 1997 at Alisthana) earlier than that indicated by the field data. It is apparent from the results that there is potential to irrigate even larger command areas than those used particularly at the tail-end tanks. The simulations show the potential use of the model in evaluating feasibility of different water issue scenarios in the process of the development of effective water management options.

Tank water balance. The model clarified the relative magnitudes of the water balance components at each tank (Fig. 9) over the simulation period. The results showed that tank inflow mainly comprised of rainfall runoff (60–85%). The rain on tank was higher (25–26%) at the two tail-end tanks compared to that at the two head-end tanks (15–17%). This is consistent with the fact that the two tail-end tanks are larger, and had greater water-spread area due to the high storage observed over most of the simulation period in comparison with the head-end tanks. The results indicated that about 14% of the total inflow to the confluence type Meegassagama tank and 4% of the total inflow to Alisthana tank were derived from the return and spill flows from the upstream tank(s). This demonstrates re-use of water from upstream catch-

ments in the immediately downstream tanks facilitated by the tank cascade system.

The significance of the tank seepage loss became clearly evident, as it accounted for 42–84% of the total inflow to the tank. The lower values were observed in the tail-end tanks, which had considerably high tank storage over most of the calibration period. In comparison, the total water release for irrigation was relatively low, and was within the range 4–22% of the total inflow. Evaporation accounted for 8–23% of the total inflow, and was higher (20–23%) at the tail-end tanks where the waterspread areas were relatively larger over the calibration period compared with the head-end tanks. The spillway discharge was estimated to be a relatively small outflow component (3–5%). The results confirm tank seepage as the main outflow component, which, in an indirect way, supplements the water requirement of vegetation in the surrounding and command areas of tanks.

5. Applicability of the modelling approach

For the efficient use of the limited water resources in regions such as the dry zone of Sri Lanka where surface water is stored in tank cascade systems, models that can adequately represent the tank water balance components need to be employed to determine feasibility of different water management scenarios. When deciding on a modelling approach applicable to a given problem, it is important to consider several factors including the scale of the problem, type of simulation, predictions required, and the accuracy of the output required. In addition, availability of the data, both for the model parameterisation and for calibration and validation, need to be taken into account.

There are a large number of hydrologic models reported in the literature (Singh, 1995), which are generally based on two alternative approaches. One of those is the physically based, distributed modelling approach, in which the effective processes are described using the principles of physics, and spatial variation of physical properties and driving variables are represented based on a suitable spatial discretisation. In contrast, the lumped-conceptual modelling approach is based on simplifications introduced through conceptualisation of the physical process.

The advantages and constraints associated with both of these modelling approaches have been discussed by several authors (Abbott et al., 1986; Bathurst and O'Connell, 1992; Beven, 1993; Abbott and Refsgaard, 1996). It has been argued that the physically based distributed parameter models, for example MIKE-SHE (Abbott et al., 1986), offer many advantages, and have the capacity to account for the spatial variability of catchment properties and physical processes. The substantial data requirements, complexity and heavy computational overhead associated with these models compromise their application at the catchment scale. The models based on the lumped-conceptual modelling approach have been successfully applied for real-time flood forecasting, and in the extension of short streamflow records using longer rainfall records by several authors. Generally, lumped-catchment models utilise a large number of parameters, for example, Stanford Watershed Model (Crawford and Linsley, 1966), and depend on the availability of long meteorological and hydrological records for calibration (Bathurst and O'Connell, 1992). In addition, due to the curve fitting aspect usually involved in the calibration of such models, physical interpretation of the model parameters is difficult (Abbott et al., 1986).

Given the constraints and the capabilities of each modelling approach, it is important to recognise their potential application in a given field situation in order to achieve the objectives of the hydrologic problem at hand. The simulation of discharge under stationary catchment conditions can be usually well addressed by the simpler lumped-conceptual type models. However, the physically based, distributed models appear as the only option capable of predicting the effects on the catchment due to land use change, and other human interferences and for simulating water quality and soil erosion in catchments (Abbott and Refsgaard, 1996). Further, the more comprehensive nature of the distributed physically based model facilitates model validation against internal state variables.

5.1. The adopted modelling approach

There is a lack of simple, yet adequate procedure to simulate water balance in cascade systems with the capability to represent temporarily and spatially

varying hydrologic processes of tank cascade systems. This has enforced key processes such as runoff generation in catchments and tank seepage and percolation loss to be approximated based on rational assumptions in simulating hydrologic behaviour of tank cascade systems. Usually in field situations with tank cascade systems (e.g. the dry-zone of Sri Lanka), standard meteorological data is the only available information. Even the data that may have been available to determine tank area and tank water volume at different tank water heights would be outdated or invalid due to on-going siltation and other problems associated with small irrigation tanks. Given the scarcity of data in such field conditions, the detailed data requirements of physically based, distributed models cannot be fulfilled. The use of a large number of parameters that require calibration and the need for long-historical records associated with lumped-conceptual models would also severely limit their potential field application in such conditions.

The simple water balance modelling approach used in the development of the *Cascade* model utilises useful aspects of both the physically based, distributed and lumped-conceptual modelling approaches. It allows complexity of the model and the data requirement to be minimal while accommodating the distributed nature of the parameters in field conditions at different tanks in the cascade. For example, runoff generation within a catchment contributing to a tank is simulated using the lumped-conceptual approach, while accommodating the distributed nature of the driving variables and the effective parameters within the cascade by allowing them to vary according to the effective field conditions at different tanks. On the other hand, the user-defined coefficients that are common to all the tanks are assigned common values as in the lumped-conceptual type models. Even though, a large number of parameters that need calibration are normally used in lumped-conceptual models, in the *Cascade* model this requirement has been limited to a minimum of two parameters (RCF and Delay). This approach has led to the development of a structurally simple model with a low calibration requirement and an input data requirement commensurate with the measurements that can be generally obtained in the field.

The Reservoir Operation Simulation (Extended) Systems model (Usgodaarachi et al., 1996), which provided the basis for initiating the development of the *Cascade* model, and the model described by Palmer et al. (1982) also provide systems for representing daily reservoir (tank) water balance. As described in Section 1, Itakura (1995) described a water balance model based on a set of tanks in the Thirappane TCS, which was extended further by Shinogi et al. (1998). The *Cascade* model differs from those systems particularly in the manner it calculates two vital tank water balance components (1) rainfall runoff and (2) tank water seepage.

5.1.1. Runoff yield calculation

The process of runoff generation in catchments is commonly approximated by the use of simple models based on the runoff coefficient method. The runoff coefficient of most linear system models is assumed to be a constant, e.g. in the water balance model described by Itakura (1995), in which the runoff coefficient was assigned an average seasonal value in estimating runoff. As pointed out by Xia et al. (1997), the runoff coefficient is by no means constant, and its variability (due to the influence of factors such as evaporation, catchment wetness and rainfall intensity) cannot be explained in terms of a simple seasonal variation. In the *Cascade* model, the non-linearity of the runoff generation process is approximated through a time-varying runoff coefficient, facilitated by the use of a modified API indicating the degree of catchment wetness. A modified API (Section 3.2.1) as against the conventional API (Linsely et al., 1988) was used for better representation of soil moisture depletion caused by infrequent but intense occurrence of rainfall events typical for this region.

5.1.2. Tank seepage estimation

In the design of irrigation headworks for small catchments in Sri Lanka, monthly seepage loss from tanks has been commonly taken as 0.5% of the tank water volume (Ponrajah, 1984). This estimate, however, is based on seepage measurements of large tanks systems constructed with modern technology and cutoff trenches (Tasumi et al., 1999), and thus would not be representative of the seepage loss in small irrigation tanks. Based on an analysis on a tank cascade system in the Anuradhapura District in

Sri Lanka, Tasumi et al. (1999) estimated the daily seepage rate as 2.4% of the tank volume, and pointed out that it can be even higher than 20% when the tank water height is small and can decrease to a very small rate close to zero at high tank water heights.

In the water balance model based on a set of tanks in the Thirappane TCS (Itakura, 1995) tank seepage and percolation loss was approximated as a linear function of the tank water surface area. Shinogi et al. (1998) extended this further by representing the tank seepage loss as a logarithmic function of the tank water surface area. The *Cascade* model calculates tank seepage and percolation loss based on functions derived from the analysis of the observed tank water loss during time periods without rainfall at the Thirappane TCS. The tank seepage variation represented by these functions (Section 3.2.2) agree with the points made by Tasumi et al. (1999) and also with the results of a water balance study at the village tank Walagambahuwa, in the dry zone of Sri Lanka (Dharmasena, 1985), where the seepage loss interpreted as a percentage of tank storage decreased as the head increased. In addition, the seepage rate variations represented by the seepage functions are consistent with the effects of different field conditions on tank water seepage. During time periods with high tank water volume, the ground water table in the surrounding area would be high. The water loss due to seepage in this condition may not be a high percentage of the tank water volume. In contrast during dry periods when the tank storage is small, water table in the surrounding area can be low, resulting in a greater head difference between tank water and the groundwater level. This can cause a significant water loss due to seepage, as dry conditions would prevail over a considerable area around the tank. Given that the tank water volume is low, water loss due to seepage can be a relatively large fraction of the tank water volume compared with the fraction of water loss due to seepage in high tank water conditions.

The model accounts for the distributed nature of the factors determining seepage by the use of seepage functions based on observations at each tank. When the model is used in other similar tank cascades where the relevant observations are not available, tank seepage can be approximated in a lumped-conceptual

manner with the use of the function representing the average seepage rates in the Thirappane TCS (Section 3.2.2).

6. Summary and conclusions

Accurate prediction of tank water availability is a vital initial step in the process of optimising usage of the limited water resources in tank cascade systems. In this study, the *Cascade* model was developed based on a simple structure and a modest input data requirement, with the aim of providing a means for determining water availability in tank cascade systems such as the Thirappane TCS.

The results of the model calibration, which involved field data representing both wet and dry conditions in the Thirappane TCS, showed a reasonable overall agreement with the field observations. The apparent discrepancies between the model and the field observations can be attributed to several contributing factors including the assumptions used in the model, inadequacy of the seepage functions, incorrect (biased) data, inter-correlation of parameters and the empirical nature of the equations. Siltation and other problems associated with village irrigation tanks can affect the water holding capacity of tanks. As a result, validity of the mathematical relationships used in the model for estimating tank capacity and tank area can change. Given the uncertainty caused by the above factors, the agreement between the observed data and the simulated results over the calibration period provides some evidence of the applicability of the methods adopted for simulating water balance of different types of tanks in the cascade system. However, as a model validation involving an independent data set could not be performed, general applicability of the methods employed and the predictive capability of the model cannot be evaluated.

Based on the results over the calibration period, several important features of the *Cascade* model that facilitated it to simulate the dynamic hydrologic components of the tank cascade system can be identified. It employs a modified runoff coefficient method for estimating runoff from rainfall. As runoff generation is a complex process

influenced by several factors, either the use of a constant or a simple seasonal value for representing the runoff coefficient variation (e.g. as used by Itakura (1995), or Kachroo and Liang (1992)) is not adequate. The method adopted by the *Cascade* model incorporates a modified API, which in an indirect way, acts as an index of the wetness of the catchment. This method allows variation of the runoff coefficient according to the catchment wetness, providing a simplified method for the representation of the non-linear runoff generation process. The simulations of the tank water storage increase in response to rainfall over the calibration period were consistent with the observed data, providing some evidence of the applicability of the modified runoff coefficient method adopted by the model for estimating rainfall runoff.

The tank seepage analysis is a highlight of this study. It led to the formulations that allow estimation of tank seepage in the model in a manner consistent with the field observations, and derivation of a function that can be used to approximate tank seepage loss in similar small irrigation tanks in the absence of the relevant field measurements.

The model simulations provided valuable insights into the water balance at each tank and displayed the relative significance of water balance components and interconnections of the tanks within the cascade system. The model indicated that over the calibration period tank seepage loss accounted up to 84% of the total inflow to the tank, proving to be a significant component in comparison with the total water release for irrigation, which was only up to 22% of total inflow.

The *Cascade* model can be further improved by using additional field observations at the Thirappane TCS and in other similar tank cascade systems by (1) refining seepage functions, (2) extending calibration and performing an independent validation based on a split-sample test, and (3) incorporating other features of other tank cascade systems such as diversion weirs and feeder canals. With such improvements, the *Cascade* model could provide a basis for the development of a useful tool for the purpose of optimising usage of the limited water resources in tank cascade systems, which are commonly found in the dry zone of Sri Lanka and in similar regions.

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